C-STGB: Real-Time Financial Surveillance Architecture & Conformal Triage Platform

(a) End-to-End Streaming Ingestion & Deep Spatio-Temporal Graph-Boost Neural Architecture

Real-Time Fast-Path Latency: 0.45 ms (P99: 0.82 ms) | 31,200 tx/s Throughput

Stage 1: Streaming Ingest
Wire & Zero-Copy Memory

SWIFT MT103 | ISO-20022 | Kafka

UTXO Blockchains (Bitcoin, Eth)

DuckDB Streaming Micro-Batches

Apache Arrow Zero-Copy Buffer

Causal Split: $t_e \leq t_{\text{eval}}$ **(No Leak)**

Ingestion Latency SLA: 0.11 ms

Stage 2: 12-D Invariants
Physical Flow & Hawkes

Mass Flow: $\Phi_{\text{flow}} = \ln(1 + \frac{A_{\text{out}}}{A_{\text{in}} + \varepsilon})$

Pass-Through Layering Detector

Hawkes Arrival: $\lambda_u(t) = \mu_u + \sum \alpha e^{-\beta \Delta t}$

Burst Acceleration Metric (μ_A, σ_A^2)

Directed Causal Taint: $s_{\text{fwd}}, s_{\text{bwd}}$

Tensor Descriptor: $\mathbf{z}_{\text{inv}} \in \mathbb{R}^{12}$

Stage 3: Temporal GNN
Tri-Band & Edge Gating

3-Band Harmonic Temporal Attention

$\tau_1 = 2.4\,\text{h}$ (burst), $\tau_2 = 24\,\text{h}$, $\tau_3 = 168\,\text{h}$

Edge Gate: $\hat{g}_{ij} = \sigma(\mathbf{w}_g^\top [\mathbf{h}_i \,\|\, \mathbf{h}_j \,\|\, \mathbf{e}_{ij}])$

Prunes 65.9% Chaff Camouflage Links

Typology GraphSMOTE on Clusters C_k

GNN Forward SLA: 0.16 ms

Stage 4: Fusion & Risk
Adaptive Head & Bayes Tau*

Adaptive Gate: $\alpha_u = \sigma(\mathbf{w}_\alpha^\top [\mathbf{h}_u \,\|\, \mathbf{z}_{\text{inv}}])$

Unified Space: $\mathbf{h}_u^* = \alpha_u \mathbf{h}_u + (1 - \alpha_u)\mathbf{z}_{\text{inv}}$

Cold-Start: 89.60% F1 for $\deg(u) \leq 2$

Cost-Sensitive Head: $\tau^* = 0.18$

15:1 Asymmetric Ratio $(C_{\text{FN}}/C_{\text{FP}})$

Minority Risk Output: $\hat{p}_u \in [0, 1]$

(b) Class-Conditional Conformal Risk Control (CRC) Triage Gate & 3-Tier Operational Dispatch

Finite-Sample Risk Bound: $P(Y \in \Gamma(X) \mid Y = y) \geq 99.0\%$ **| 0.00% Empty Sets**

Holdout Calibration

Holdout Split: $\mathcal{D}_{\text{cal}}$

$n_0 = 3,306, \;\; n_1 = 372$

Non-Conformity Scores:

$s_i^{(1)} = 1 - \hat{p}_i$

$s_i^{(0)} = \hat{p}_i$

Target Coverage: 99.0%

Class-Cond.

CRC Triage

$1 - \alpha \geq 99\%$

$\hat{q}^{(0)}, \hat{q}^{(1)}$

Tier 1: High-Risk Immediate Quarantine
$\Gamma(u) = \{1\}$ **(Decisive Illicit)** **18 alerts / 10,000 tx (0.18%)**

Immediate Quarantine Hold | Automated Wire Freeze

Generates FinCEN Form 111 SAR Dossier with GNNExplainer Attribution Subgraph & SHA-256 Merkle Audit Receipt

Tier 2: Compliance Officer Review Desk
49 alerts / 10,000 tx (0.49%) $\Gamma(u) = \{0, 1\}$ **(Abstention Boundary)** **Workload Reduction: 99.51%**

Human-in-the-Loop 4-Eyes Compliance Review Queue

Ambiguous transactions routed to senior examiners; cuts false alarms by 85.7%--95.8% vs rules; feeds Active Learning loop

Tier 3: Automated Straight-Through Clearing
$\Gamma(u) = \{0\}$ **(Decisive Benign)** **9,933 transactions / 10,000 tx (99.33%)**

Automated Wire-Speed Straight-Through Processing (STP)

Instantaneous clearance under 0.45 ms SLA with mathematically bounded selective risk (residual illicit leakage < 0.13%)

REAL-TIME PRODUCTION SLAS | STATISTICAL SAFETY GUARANTEES | REGULATORY GOVERNANCE

Latency Performance	Workload Efficiency	Statistical Safety	Regulatory Governance
0.45 ms Fast-Path	**-99.51% Review Cut**	$\geq 99.0\%$ **Coverage**	**SR 26-2 & OCC Ready**
P99: 0.82 ms | 31,200 tx/s Throughput	Only 49 of 10,000 transactions reviewed	Finite-sample exchangeability bound	Zero-hallucination graph evidence
Strict compliance with <10 ms SLA	85.7%--95.8% fewer false alerts vs rules	0.00% empty sets | Avg set size: 1.005	Cryptographic SHA-256 Merkle audit trail